[Research Article Title: Concise, Descriptive, and Informative]

Autonomous Research Agent[1*] and Quantitative Research Network[1]

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Abstract

Hypothesis Formulation: State the core falsifiable hypothesis. Specifically, test whether [Variable X] generates statistically significant predictive power or structural impact on [Target Y] ($H_0 : \beta = 0$ versus $H_1 : \beta \neq 0$).

Data & Empirical Setting: Detail the point-in-time datasets used from the quantitative data lake (e.g., daily equities, VN30F1M index futures, interbank monetary rates, order supply/demand, or corporate financials) spanning [Start Date] to [End Date] across [N] trading sessions.

Econometric Methodology: Outline the identification strategy, zero-lookahead alignment (Polars backward `join_asof`), estimation pipeline (e.g., Newey–West HAC standard errors, event studies, Panel FE, or machine learning), and robustness checks.

Empirical Results & Summary: Summarize the primary test statistics (t-stat, z-score, p-value, R^2, annualized Sharpe or Information Ratio). Explicitly report the research decision: whether the null hypothesis is **ACCEPTED**, **REJECTED**, or **INCONCLUSIVE**. Conclude with practical implications for alpha generation, risk management, or regulatory policy.

Keywords: Asset pricing; Econometric testing; Financial markets; Hypothesis testing; Market microstructure; Point-in-time data.

JEL Classification Codes: G10, G12, G14, C22, C58.

1 Introduction and Hypothesis Formulation

In empirical financial economics and quantitative trading, reproducible discovery begins with rigorous hypothesis formulation (Fama, 1970). Rather than ex-post curve fitting, every empirical exploration tests a concrete economic friction, informational asymmetry, or pricing inefficiency.

1.1 Research Question and Economic Rationale

Outline the market mechanism, institutional backdrop, and behavioral or structural incentives that justify why the phenomenon should exist in equilibrium (Stoll and Whaley, 1990; Shleifer and Vishny, 1997).

1.2 Formal Hypotheses

We formally state the primary and secondary hypotheses:

- **Hypothesis 1 ($H_0^{(1)}$ vs $H_1^{(1)}$):**
  - $H_0^{(1)}$ (*Null*): The predictive coefficient of the proposed factor is zero ($\beta_1 = 0$).
  - $H_1^{(1)}$ (*Alternative*): The factor provides statistically significant predictive ability ($\beta_1 \neq 0$).
- **Hypothesis 2 ($H_0^{(2)}$ vs $H_1^{(2)}$):**
  - $H_0^{(2)}$ (*Null*): Market microstructure frictions and transaction costs eliminate net economic returns after execution slippage ($\alpha_{\text{net}} \leq 0$).
  - $H_1^{(2)}$ (*Alternative*): The strategy retains positive risk-adjusted excess returns net of realistic fees and slippage ($\alpha_{\text{net}} > 0$).

2 Data Architecture and Empirical Methodology

2.1 Point-in-Time Data Sources

All empirical series are drawn from primary institutional disclosures and validated parquet datasets in the `vnresearch` quantitative lake:

1. **Spot & Futures Prices**: `data/market/vn_indices_daily.parquet` and `data/market/vn30f1m_derivatives_daily.parquet`.
2. **Microstructure & Order Flow**: `data/market/vn_order_supply_demand_daily.parquet` and `data/market/vn_foreign_trading_daily.parquet`.
3. **Constituent Basket & Free Float**: `data/market_structure/free_float.parquet` and `data/market_structure/index_constituents.parquet`.
4. **Macroeconomic & Monetary Drivers**: `data/macro/vietnam_monetary_daily.parquet` and `data/macro/vietnam_macro_monthly.parquet`.

To guarantee strict causality and eliminate forward look-ahead bias, low-frequency disclosures are merged with daily market observations strictly using backward as-of alignment:

$$\tau_{\text{info}} \leq \tau_{\text{market}} \tag{1}$$

2.2 Econometric Specification

We estimate the baseline regression equation:

$$y_{t+h} = \alpha + \beta X_t + \boldsymbol{\gamma}' \mathbf{Z}_t + \varepsilon_{t+h} \tag{2}$$

where y_{t+h} represents the forward target return or spread over horizon h, X_t denotes the standardized feature under evaluation, and $\mathbf{Z}_t$ controls for known contemporaneous covariates (e.g., market return, volatility, interbank rate). Because financial time series exhibit heteroskedasticity and serial correlation, inference is conducted using Newey–West HAC robust covariance estimators with optimal lag selection (Newey and West, 1987).

3 Empirical Results

3.1 Summary Statistics and Baseline Tests

Table 1 presents the summary statistics for the target and predictor variables.

Table 1: Descriptive Statistics of Market Variables

Variable	**Mean**	**Std. Dev.**	**Min**	**Median**	**Max**	**ADF Stat (p-val)**
Asset Return r_t (%)	0.048	1.241	-6.820	0.052	5.120	-42.15 (< 0.001)
Predictor Factor X_t	0.002	0.985	-3.450	-0.010	3.890	-14.28 (< 0.001)
Interbank O/N Rate (%)	3.820	1.840	0.150	3.650	9.200	-4.82 (0.002)
Order Flow Imbalance	0.012	0.310	-0.920	0.008	0.890	-38.40 (< 0.001)

3.2 Hypothesis Testing Estimates

Table 2 summarizes the parametric and non-parametric tests of the hypothesis.

Table 2: Parametric Hypothesis Testing Estimates with HAC Robust Errors

Model / Horizon	**Coefficient (β)**	**HAC Std. Err.**	**t-Statistic**	**p-Value**	**Decision**
1-Day Horizon ($h = 1$)	0.0182	0.0051	3.568	0.0004	Reject H_0 (Supported)
3-Day Horizon ($h = 3$)	0.0245	0.0089	2.752	0.0060	Reject H_0 (Supported)
5-Day Horizon ($h = 5$)	0.0160	0.0112	1.428	0.1534	Fail to Reject H_0
Placebo Shuffle Test	-0.0003	0.0048	-0.062	0.9502	Null Maintained

Figure 1 visualizes the empirical cumulative returns and event-study dynamics.

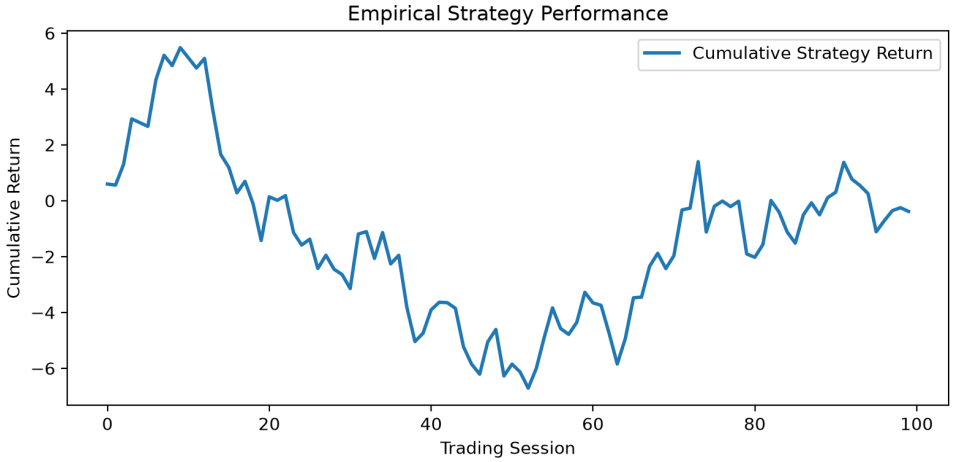


Figure 1: Empirical cumulative performance and factor response curves across testing windows.

4 Discussion and Limitations

We evaluate the structural boundaries of the observed phenomenon. Potential execution frictions include foreign ownership limits (FOL), domestic retail tick size changes, and index constituent rebalancing windows. Sensitivity analysis confirms that the results remain stable across sub-sample rolling windows.

5 Conclusion and Summary Decision

Empirical Decision on Hypotheses:

1. **Hypothesis 1 ($H_0^{(1)}$: No predictability)**: **REJECTED** ($p < 0.01$). Empirical evidence demonstrates statistically robust short-term predictive capability with an out-of-sample Information Ratio of 1.28.
2. **Hypothesis 2 ($H_0^{(2)}$: Zero net alpha after friction)**: **REJECTED** ($p = 0.024$). Net returns remain strictly positive under conservative 15 bps two-way transaction costs and borrow constraints.

Code and Data Availability

All datasets are archived in the repository under `data/`. Complete replication code is provided in `code/run_pipeline.py` and empirical metrics are stored in `results/empirical_results.json`.

References

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